JEE-Main-21-01-2026 (Memory Based)
[MORNING SHIFT]
Physics

Question: Two ideal springs S_1 and S_2 are connected in parallel between a rigid support and a massless plate. Their spring constants are measured as
$k_1 = (a \pm \triangle a)$, $k_2 = (b \pm \triangle b)$
where $\triangle a$ and $\triangle b$ represent maximum absolute errors. The effective spring constant of the combination and the maximum absolute error in it are:

Options:
(a) $((a + b) \pm (\triangle a - \triangle b))$
(b) $((a + b) \pm (\triangle a + \triangle b))$
(c) $((ab)/(a + b) \pm (\triangle a + \triangle b))$
(d) $((ab)/(a + b) \pm (\triangle a - \triangle b))$

Answer: (b)

Solution:

For springs in parallel:

$$k_{eq} = k_1 + k_2$$

Given:

$$k_1 = (a \pm \Delta a), \quad k_2 = (b \pm \Delta b)$$

So,

$$k_{eq} = (a + b)$$

For maximum absolute error in a sum:

$$\Delta k_{eq} = \Delta a + \Delta b$$

Hence,

$$\boxed{k_{eq} = ((a + b) \pm (\Delta a + \Delta b))}$$

Correct option: (B)

Question: Find the moment of inertia of a T shaped. Object formed by joining two rods of equal length about an axis which is passing through the junction and which is perpendicular to the object's plane. Consider mass of rod as M & length as L.

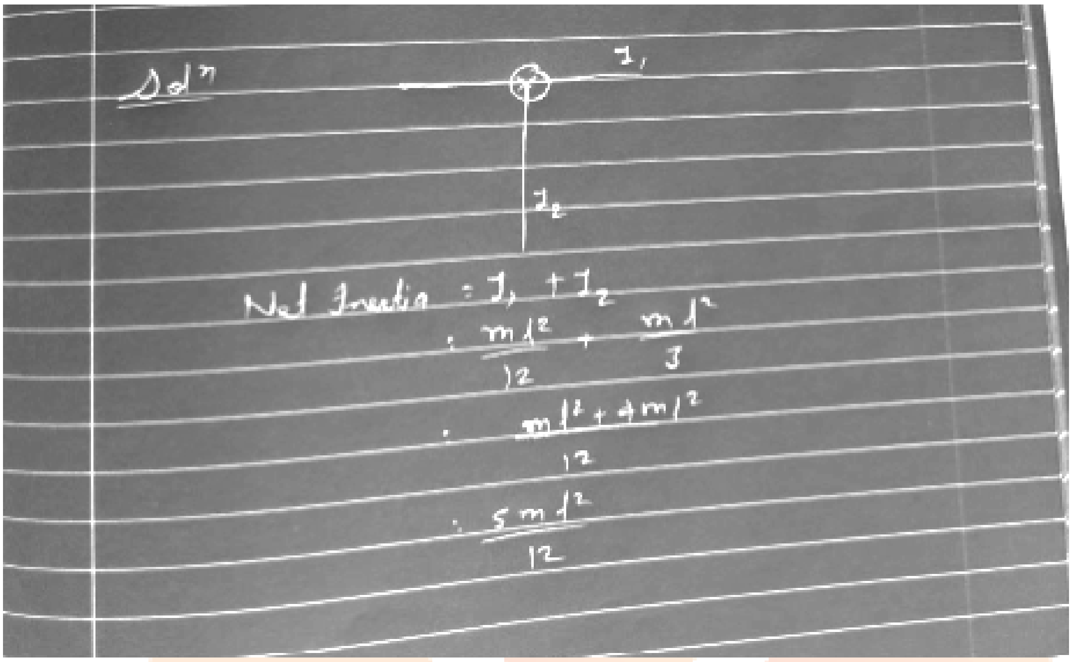
Options:
(a) $\frac{5ml^2}{12}$
(b) $\frac{3ml^2}{7}$

(c) $\frac{3ml^2}{5}$

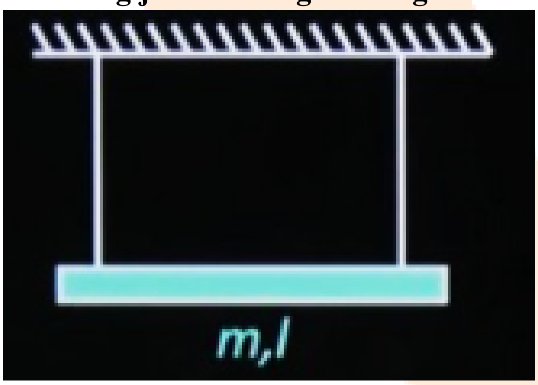
(d) ml^2

Answer: (a)

Solution:



Question: A rod of mass m and length l is attached to two ideal strings. Find tension in left string just after right string is cut.



Options:

(a) $\frac{mg}{2}$

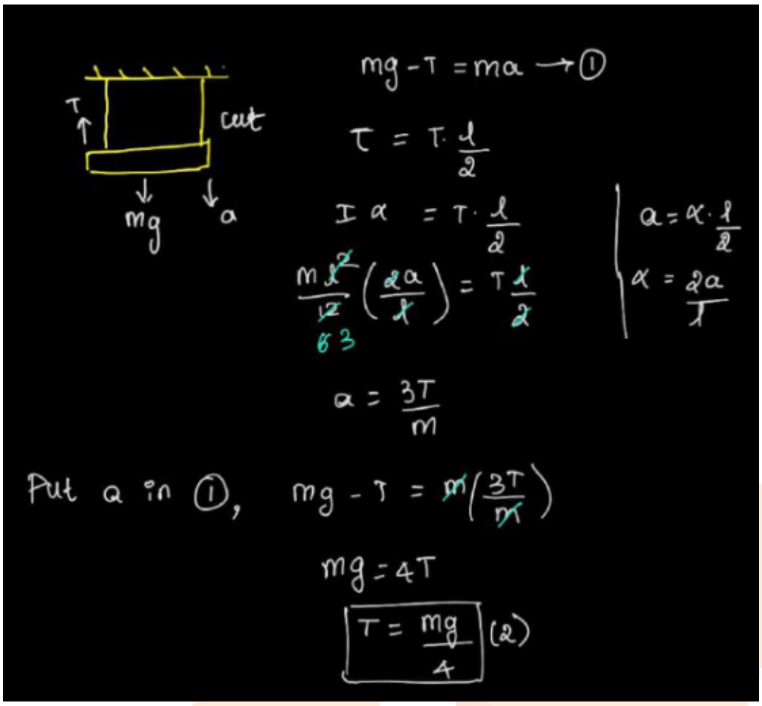
(b) $\frac{mg}{4}$

(c) $\frac{2}{3}mg$

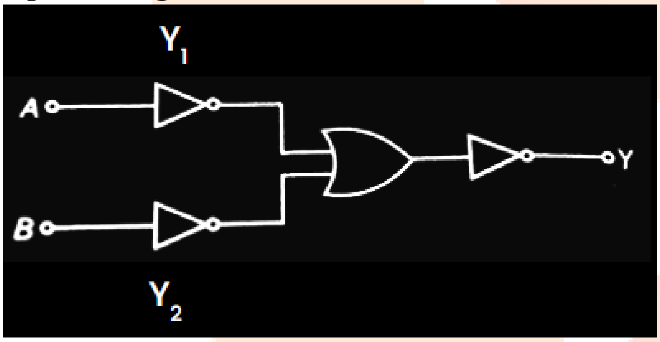
(d) $\frac{mg}{5}$

Answer: (b)

Solution:



Question: Figure shows a logic gate circuit. Identify the GATE which the circuit is representing.



Options:

(a) XOR

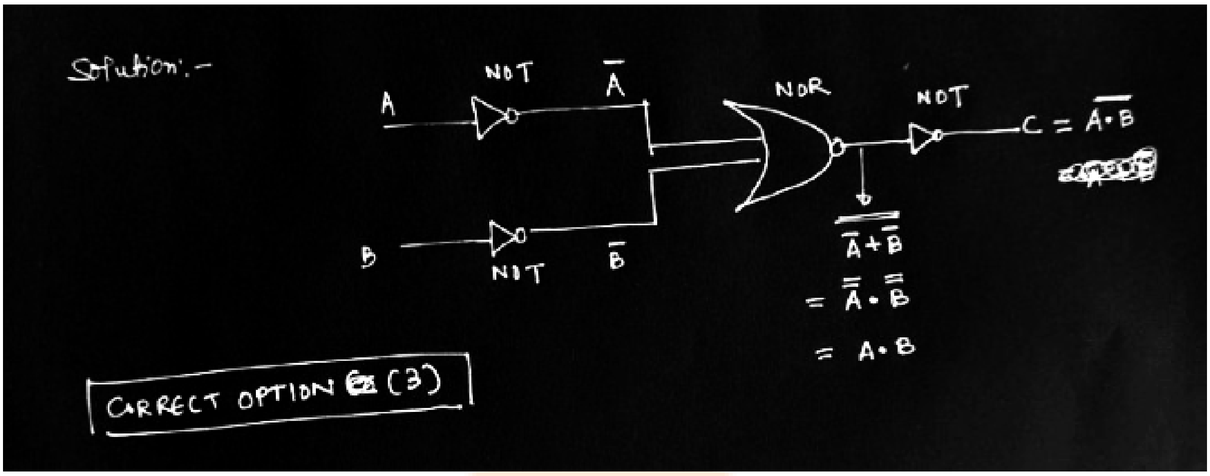
(b) NOR

(c) NAND

(d) OR

Answer: (c)

Solution:



Question: In a Certain Fluid flow the following relation holds

$$\frac{A}{B}\text{ if }\left(P+\frac{At^2}{B}\right)+\frac{1}{2}\rho V^2=\text{constant},$$ **where P is pressure, ρ is density, V is speed.**

Dimension of A/B are?

Options:

(a) $ML^{1}T^{-4}$

(b) $ML^{-1}T^{-4}$

(c) $ML^{2}T^{-4}$

(d) $ML^{-1}T^{-2}$

Answer: (b)

$$[P]=\left[\frac{At^2}{B}\right]$$

$$\left[\frac{A}{B}\right]=\frac{[\rho]}{[t^2]}=\frac{[M^1L^{-1}T^{-2}]}{[T^2]}=[M^1L^{-1}T^{-4}]$$

Question: An ideal gas sample contains 10 moles of O_2. Its molar heat capacity at constant pressure is 7 cal mol^{-1} K^{-1} and R is 2 cal mol^{-1} K^{-1}. The internal energy of the gas at temperature T K is:

Options:

(a) 50T cal

(b) 50T cal

(c) 50T cal

(d) 50T cal

Answer: (a)

Solution:

Concise Solution

For an ideal gas:

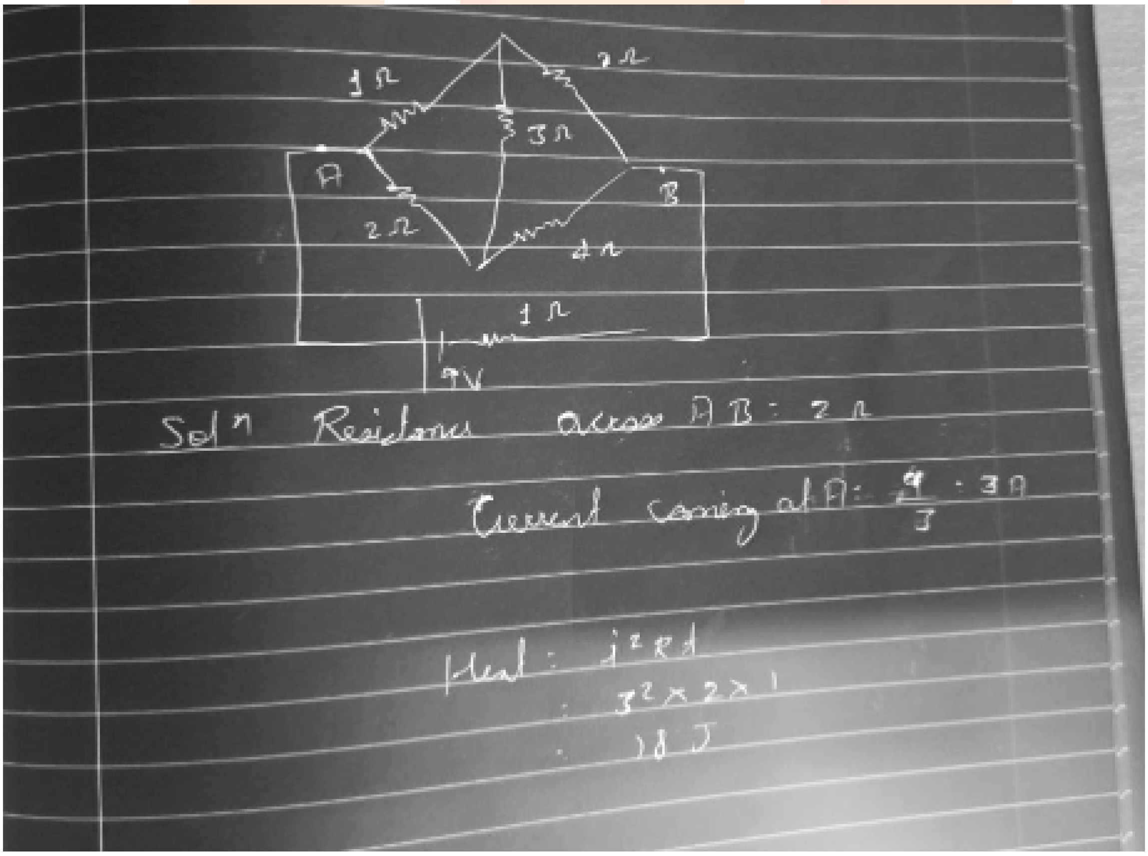
$$C_v = C_p - R = 7 - 2 = 5 \text{ cal mol}^{-1}\text{K}^{-1}$$

Internal energy:

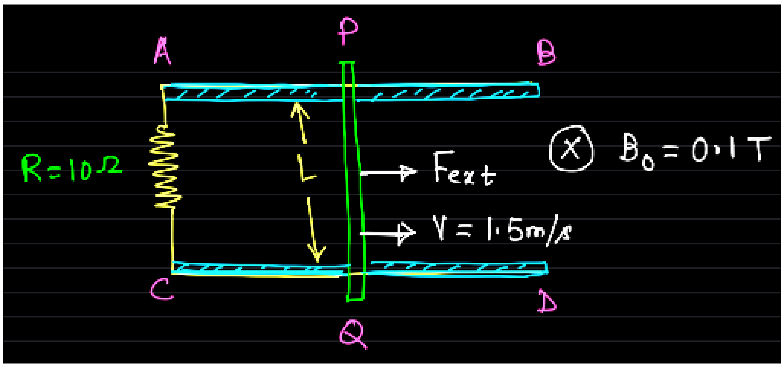
$$U = nC_vT = 10 \times 5 \times T = 50T \text{ cal}$$

Correct option: (A) $50T$ cal

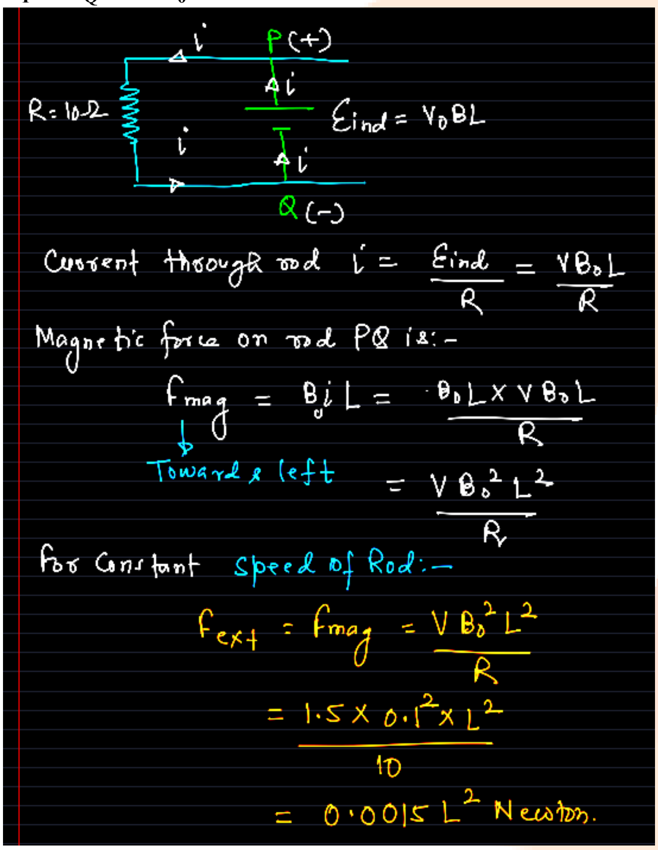
Question: Find the heat dissipated across the wheatstone bridge circuit (about AB) in the given circuit for a time interval of 1 second.
Options:
(a) 18 J
(b) 54 J
(c) 27 J
(d) 16 J
Answer: (a)
Solution:



Question: Consider two smooth parallel conducting rails AB & CD bridged by a 10 Ω resistor. A Conducting rod PQ Length L is placed over the two rails perpendicular as shown. The region consists of a uniform magnetic field of intensity 0.1 T perpendicular to the plane of figure & going inside. A constant force F_{ext} is required to pull the rod towards right so that it acquires a constant speed V = 1.5 m/A. Find the valve of F_{ext}.



Answer: (0.0015 L^2 N)
Solution:
Motional emf across end of rod PQ is:-
V_P - V_Q = VB_0L = Find



Question: Two strings of length l_A and l_B having Linear mass densities of μ_A and μ_B. Tension in both Strings is 500 N. A transverse pulse is been generated in both of them. Find ratio of time taken by pulse in string.

Answer: ($\therefore \frac{t_A}{t_B} = \frac{L_A}{L_B}\sqrt{\frac{\mu_A}{\mu_B}}$)
Solution:

Ans:- $\frac{t_A}{t_B} = ?$ $\qquad t_A = \frac{L_A}{v_A}$; $t_B = \frac{L_B}{v_B}$

$v_A = \sqrt{\frac{T}{\mu_A}}$; $v_B = \sqrt{\frac{T}{\mu_B}}$

$\frac{L_A}{v_A} \times \frac{v_B}{L_B}$

$\frac{v_A}{v_B} = \sqrt{\frac{\mu_B}{\mu_A}}$

$$\boxed{\therefore \frac{t_A}{t_B} = \frac{L_A}{L_B}\sqrt{\frac{\mu_A}{\mu_B}}}$$ Ans

Question: A particle of mass m = 4 kg moves in the xy-plane under a time-dependent force $\overrightarrow{F}(t) = 4t^3\hat{i} - 3t^2\hat{j}$ (SI units).

At t = 0, the particle is at the origin and at rest. The velocity $\overrightarrow{v}$ and displacement $\Delta\overrightarrow{r}$ at t = 2s are:

Options:

(a) $\overrightarrow{v} = \left(4\hat{i} - 2\hat{j}\right)ms^{-1}, \Delta\overrightarrow{r} = \left(\frac{8}{5}\hat{i} - \hat{j}\right)m$

(b) $\overrightarrow{v} = \left(2\hat{i} - 4\hat{j}\right)ms^{-1}, \Delta\overrightarrow{r} = \left(\frac{8}{5}\hat{i} - \hat{j}\right)m$

(c) $\overrightarrow{v} = \left(4\hat{i} + 2\hat{j}\right)ms^{-1}, \Delta\overrightarrow{r} = \left(\frac{8}{5}\hat{i} + \hat{j}\right)m$

(d) $\overrightarrow{v} = \left(8\hat{i} - 4\hat{j}\right)ms^{-1}, \Delta\overrightarrow{r} = \left(\frac{16}{5}\hat{i} - 2\hat{j}\right)m$

Answer: (a)

Solution:

$$\vec{a} = \frac{\vec{F}}{m} = t^3\hat{i} - \frac{3}{4}t^2\hat{j}$$

Using $\vec{v}(0) = 0$:

$$v_x = \int t^3 dt = \frac{t^4}{4}, \qquad v_y = \int -\frac{3}{4}t^2 dt = -\frac{t^3}{4}$$

$$\Rightarrow \vec{v}(2) = (4\hat{i} - 2\hat{j})\ \mathrm{m\,s^{-1}}$$

Using $\Delta\vec{r} = \int \vec{v}\,dt$ and $\vec{r}(0) = 0$:

$$\Delta x = \int \frac{t^4}{4} dt = \frac{t^5}{20}, \qquad \Delta y = \int -\frac{t^3}{4} dt = -\frac{t^4}{16}$$

$$\Rightarrow \Delta\vec{r}(2) = \left(\frac{8}{5}\hat{i} - \hat{j}\right)\ \mathrm{m}$$

Question: Find the work done in moving a satellite from an orbit whose radius is $R_\in$ to an orbit where radius is $3/2R_\in$, where mass of planet is M & mass of satellite is m and $R_\in$ is radius of planet.

Options:

(a) $\frac{GMm}{6R_\in}$

(b) $\frac{GMm}{5R_\in}$

(c) $-\frac{GMm}{R_\in}$

(d) $\frac{GMm}{R_\in}$

Answer: (a)

Solution:

Soln W = Change in total Energy

$= E_f - E_i$

$= \frac{-GMm}{2 \times \frac{3}{2} R_E} - \left(\frac{-GMm}{2 \times R_E}\right)$

$= -\frac{GMm}{3R_E} + \frac{GMm}{2R_E}$

$= \frac{-2GMm + 3GMm}{6R_E}$

$= \frac{GMm}{6R_E}$

Question: A fixed charge 'Q' of 1 c at origin. The work done in moving a charge of 2 μc from Point A (4, 4, 2) to point B (2, 2, 1) is _______ J.
Answer: (Ans : 3000 J)
Solution:

Ans:-

$$U = \frac{kq_1q_2}{r}$$

$$U_i = \frac{k(1)(2\times10^{-6})}{\sqrt{4^2+4^2+2^2}} = \frac{k(2\times10^{-6})}{6}$$

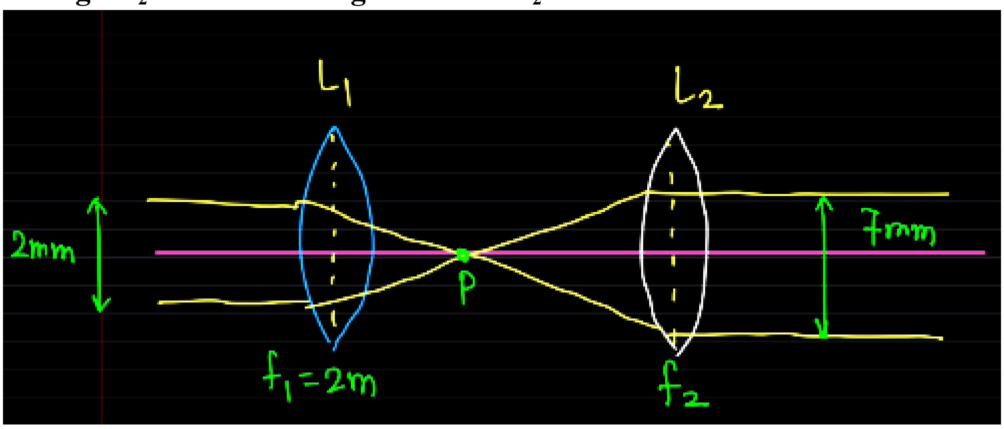
$$U_f = \frac{k(2\times10^{-6})}{\sqrt{2^2+2^2+1^2}} = \frac{k(2\times10^{-6})}{3}$$

$$W = U_f - U_i = k(2\times10^{-6})\left[\frac{1}{3}-\frac{1}{6}\right]$$

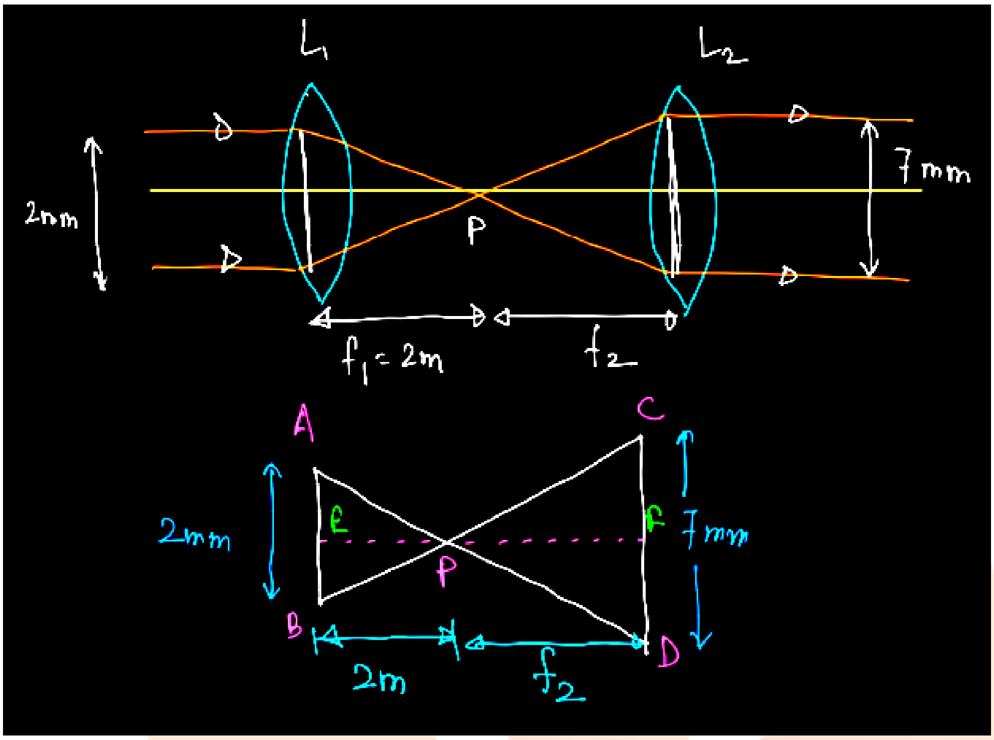
$$W = 18\times10^{3}\left[\frac{1}{6}\right] = 3\times10^{3}$$

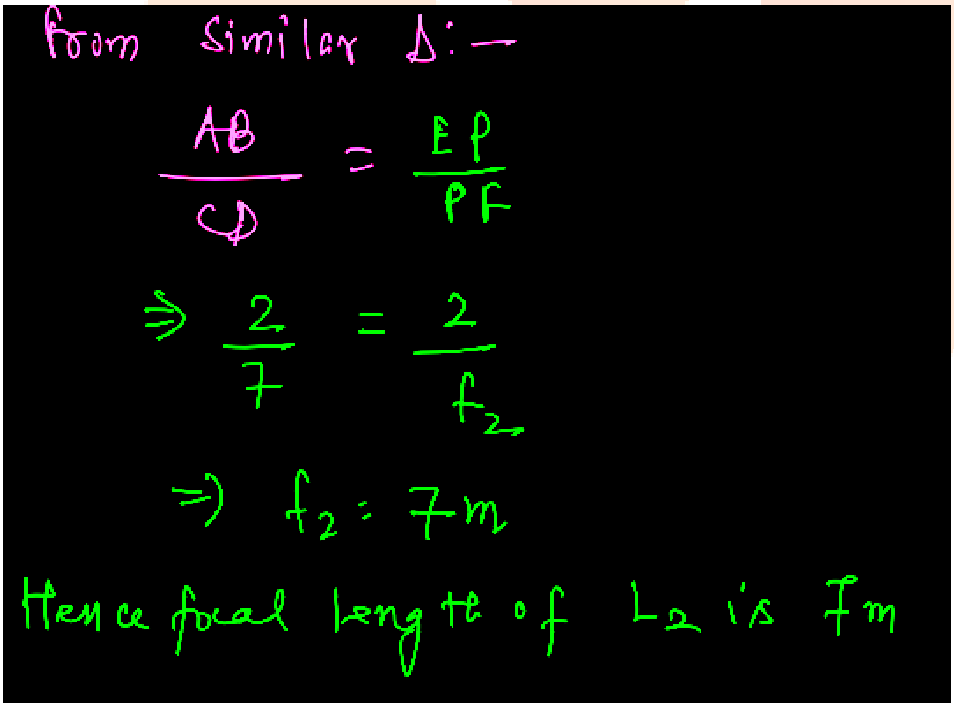
$$\boxed{W = 3000\ J}$$

Question: Consider a system of two coaxial Convex lenses L_1 & L_2 having focal length f_1 = 2 mm & f_2 (unknown) respectively. A parallel beam of Light having diameter 2 mm is incident on L_1 & it emerges out at parallel beam of diameter 7 mm after refraction through L_2. Find focal length of Lens L_2.



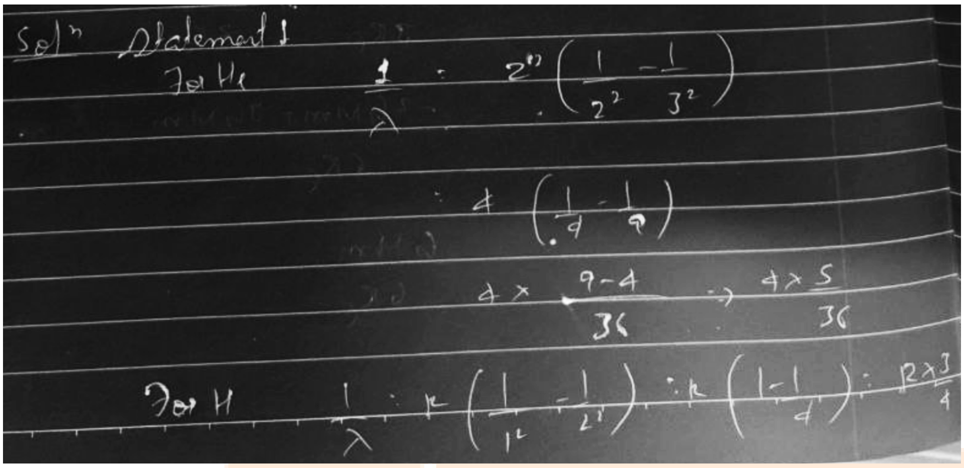
Answer: (7 m)
Solution:
Point P is the Common focus of L_1 & L_2





Question: Statement 1:- During transition of on electron in an Helium ion (He^{+}) from orbit 3 to 2, the wavelength of the photon released in equal to the wavelength of the photon which is released in the transition from 2 to 1 in hydrogen.
Statement 2:- When a H_2 molecules dissociated into respective atom, energy is released.
Options:
(a) Statement 1 & 2 both are correct

(b) Statement 1 is correct but 2 is incorrect
(c) Statement 1 is incorrect but 2 is correct
(d) Both are incorrect
Answer: (c)
Solution:


Solⁿ Statement 1

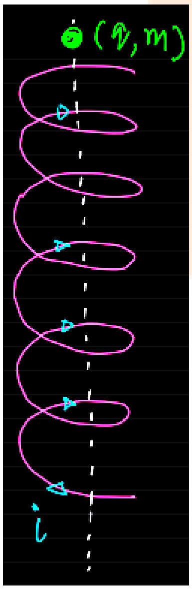
For He $\frac{1}{\lambda} = z^2 R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$= 4\left(\frac{1}{4} - \frac{1}{9}\right)$

$4 \times \frac{9-4}{36} \rightarrow \frac{4 \times 5}{36}$

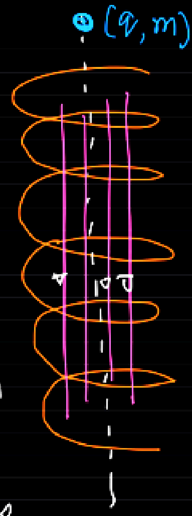
For H $\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R\left(1 - \frac{1}{4}\right) = \frac{R \times 3}{4}$

Question: Figure shows a long Current Carrying solenoid. A charge partide having charge q & mass m is released from one end of the solenoid in vertical plane as shown. If "a" denotes acceleration of charge while passing the solenoid & “g" denotes acceleration due to gravity then choose the Correct option.



Options:
(a) a = g
(b) a = 0
(c) 0 < a < g
(d) a > g
Answer: (a)
Solution:

- Inside the long solenoid, magnetic field is uniform & parallel to axis. Since charge is released from rest initially no magnetic force can act as $V = 0$. So under gravity, charge will be accelerated with acc^n g. Once it acquires velocity; magnetic force is still zero as $\vec{V}$ & $\vec{B}$ are parallel by the equation



$$F = qVB \sin 0^\circ = 0$$

Hence charge will experience only gravitational force hence its acc^n will be equal to g.

Question: A conducting circular loop of area 1m^2 is placed perpendicular to a magnetic field which varies as B = sin(100t) If resistance of loop is 100 Ω then the average thermal energy dissipated in the loop in one period is

Options:

(a) 2π

(b) π

(c) π^2

(d) $\pi/2$

Answer: (b)

Solution:

Ans: $\phi = BA$

$= (\sin 100t) \times 1$

$\phi = \sin 100t$

$e = -\dfrac{d\phi}{dt} = 100 \cos 100t$

$I = \dfrac{e}{R} = \dfrac{\cancel{100} \cos 100t}{\cancel{100}} = \cos 100t$

Power, $P = I^2 R$

$= \cos^2(100t) \times 100 = 100 \cos^2 100t$

$\langle \cos^2 \rangle = \dfrac{1}{2}$ | $P_{avg} = 100 \times \dfrac{1}{2} = 50$

Time Period, $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{100} = \dfrac{\pi}{50}$

Average thermal energy, $E = P_{avg} \times T$

$= 50 \times \dfrac{\pi}{50}$

$\boxed{E = \pi}$ Ans

Question: In a double slit experiment the distance between the slits is 0.1 cm & the screen is placed at a distance of 50 cm from the slits. If one slit is covered with a glass slab of refractive index 1.5 and thickness 't' then central bright fringe shift by 0.2 cm. Find the value of 't'.

Options:

(a) 0.0008 cm

(b) 0.0001 cm

(c) 0.0006 cm

(d) 0.0005 cm

Answer: (a)

Solution:

Shift $= (\mu - 1)\, t\, \dfrac{D}{d}$

$0.2 = 0.5 \times t \times \dfrac{50}{0.1}$

$\dfrac{0.2}{250} = t$

Question: An α-particle having K.E. = 7.7 MeV is approaching fixed gold nucleus (Z = 79). Find distance of closest approach.

Options:

(a) 20 fm
(b) 25 fm
(c) 30 fm
(d) 15 fm
Answer: (c)
Solution:

Distance of closest approach is:-

$$r_{min} = \frac{2kZe^2}{KE_i}$$

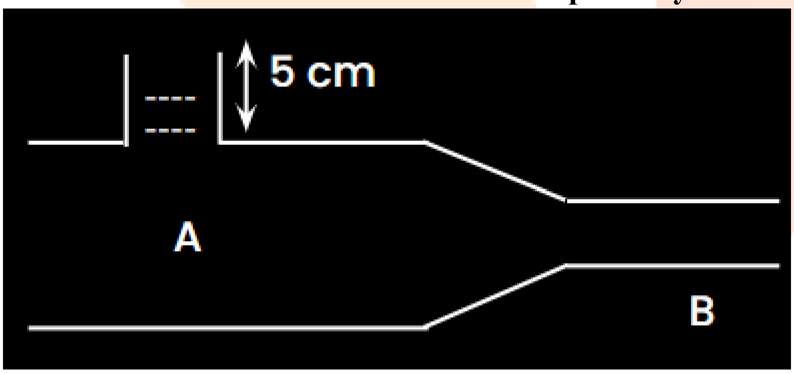
$$= \frac{2 \times 9 \times 10^9 \times 79 \times (1.6 \times 10^{-19})^2}{7.7 \times 10^6 \times 1.6 \times 10^{-19}}$$

$$= 295.48 \times 10^9 \times 10^{-6} \times 10^{-19} \times 10^9 \text{ nm}$$

$$= 295\ 48 \times 10^{-7} \text{ nm}$$

No option matching.

Question: Find volume flow rate in the given venturi meter.
Cross sectional area at A & B is A & a respectively and A/a = 2; 4A : √3 m².



Options:
(a) 0.25 m^3/s
(b) 1 m^3/s
(c) 2 m^3/s
(d) 3 m^3/s
Answer: (a)
Solution:

Sol^n

$$\rho g h + \frac{1}{2}\rho v_A^2 = \frac{1}{2}\rho v_B^2$$

$$\rho g h = \frac{1}{2}\rho (v_B^2 - v_A^2)$$

$$A v_A = a v_B$$

$$2 v_A = v_B$$

$$= \frac{1}{2} \times (4v_A^2 - v_A^2)$$

$$gh = \frac{1}{2} \times 3v_A^2$$

$$\sqrt{\frac{2gh}{3}} = v_A$$

$$R = v_B \times \frac{\sqrt{3}}{4}$$

Question: Electric field equation of a plane electromagnetic wave is given as:

$$\vec{E} = 69\sin(\omega t - kx)\hat{j}.$$

Find the direction of propagation of magnetic field vector.

Options:

(a) Along +ve z-axis

(b) Along -ve z-axis

(c) Along -ve x-axis

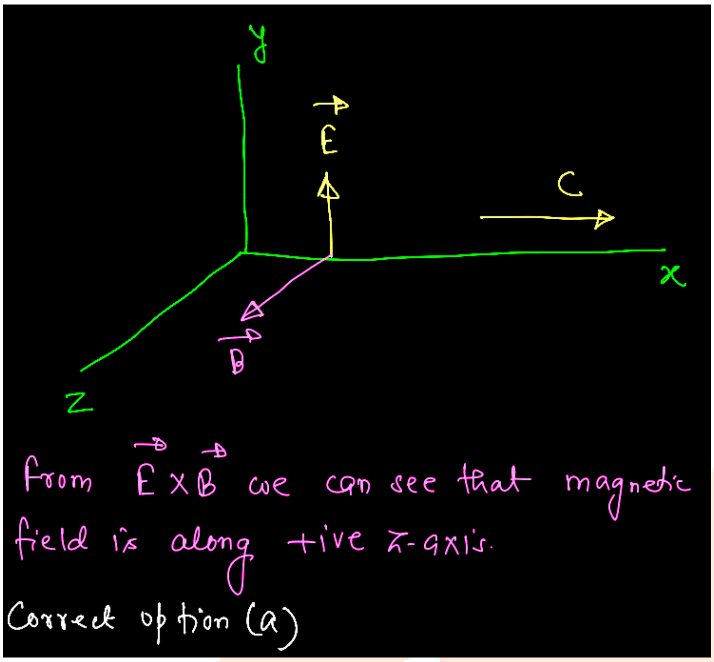
(d) Along +ve y-axis

Answer: (a)

Solution:

EM wave is propagating along +ive x-axis at ωt & kx have opposite sign.

Electric field vector is oscillating along +ive y-axis.



Question: Two rods of equal length of 60 cm each are joined together end to end. Coefficient of linear expansion of rods are 24 × 10^{-6} °C^{-1} and 1.2 × 10^{-5} °C^{-1}. Their temperatures are same and equal to 30°C, which is increased to 100°C. Find final length of combination (in cm).

Options:

(a) 120.1321

(b) 120.1123

(c) 120.1512

(d) 120.1084

Answer: (c)

Solution:

$$\Delta L_{total} = \Delta L_1 + \Delta L_2$$

$$= L_1 \alpha_1 \Delta T + L_2 \alpha_2 \Delta T$$

$$= 70 \left[60 \times 24 \times 10^{-6} + 60 \times 1.2 \times 10^{-5}\right]$$

$$= 70 \times 60 \times 10^{-6} \left[24 + 12\right]$$

$$\Delta L_{total} = 0.1512 \text{ cm}$$

$$\Delta T = T_2 - T_1$$

$$= 100 - 30$$

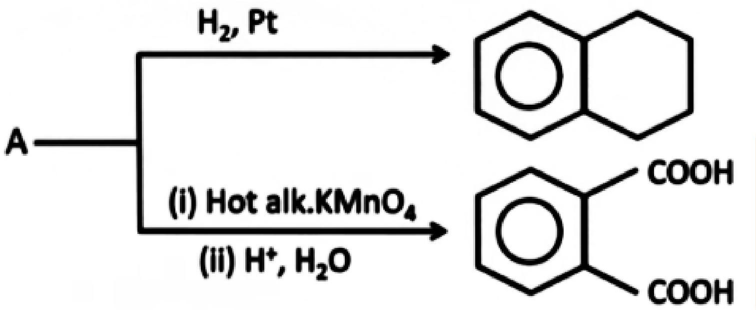
$$= 70$$

$$\therefore L_{final} = 120 + 0.1512 = 120.1512 \text{ cm}$$

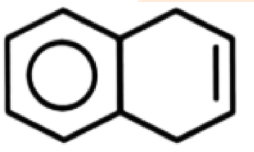
$$\boxed{L_{final} = 120.1512 \text{ cm}}$$ Ans

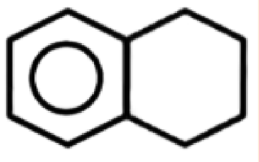
JEE-Main-21-01-2026 (Memory Based)
[MORNING SHIFT]
Chemistry

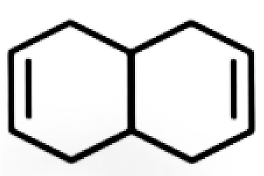
Question:

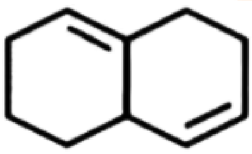


Options:

(a) 

(b) 

(c) 

(d) 

Answer: (a)

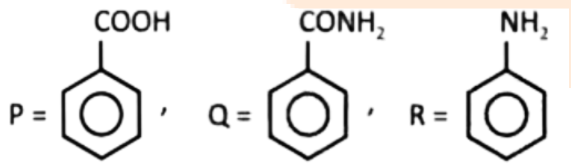
Question: For two chemical reactions A and B, if the difference between their activation energy is 20 kJ at 300 K($R = 8.3\ J\ K^{-1}\ mol^{-1}$). Determine $ln\frac{k_2}{k_1}$

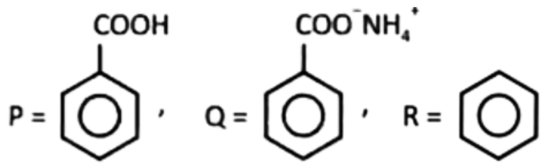
Options:
(a) 9
(b) 8
(c) 5
(d) 2
Answer: (b)

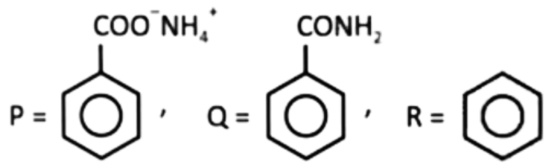
Question: In Sulphur estimation 0.7 g of an organic compound gives 1 g $BaSO_4$ in Carius method. What is the % of Sulphur in compound?
Options:
(a) 19.61
(b) 23.85
(c) 27.93
(d) 14.57
Answer: (a)

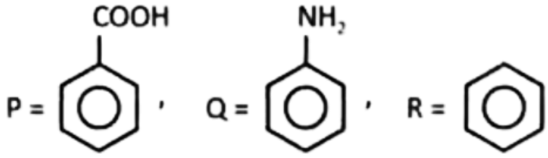
Question: Which of the following is the correct order with respect to the property indicated?
Options:
(a) Cl > F (Ionisation energy)
(b) $K_2O > Na_2O > Al_2O_3$ (Basic nature)
(c) K > Na > Al > Mg (Metallic character)
(d) None of these
Answer: (b)

Question: Given below are two statements.
Statement-I: Arginine and Tryptophan are essential amino acids.
Statement-II: Glycine does not have any chiral carbon.
In the light of the above statements, which is the correct option.
Options:
(a) Both Statement-I and Statement-II are correct
(b) Both Statement-I and Statement-II are incorrect
(c) Statement-I is correct and Statement-II is incorrect
(d) Statement-I is incorrect and Statement-II is correct
Answer: (a)

Question: Observe the following reaction sequence:
$$(P) \xrightarrow[\Delta]{NH_3} (Q) \xrightarrow{Br_2, KOH} (R)$$
Which of the following is the correct structure for P, Q and R?
Options:
(a) 

(b) 

(c) 

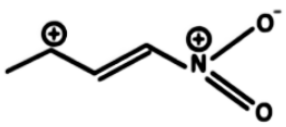
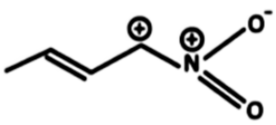
(d) 

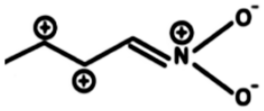
Answer: (a)

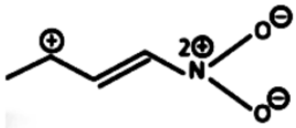
Question: In the following reaction, $MnO_4^{2-} \xrightarrow{H^+}$
Manganate ion undergoes ion undergoes disproportionation to form
Options:
(a) MnO_2, MnO_4^-
(b) MnO, MnO_2
(c) MnO_2, MnO
(d) MnO_4^-, MnO
Answer: (a)

Question: 80 mL of organic compound is mixed with 264 mL O_2 and ignited. It gives 224 mL of gaseous mixture at NTP. After passing KOH 64 mL of gas remains. The organic compound is
Options:
(a) C_2H_4
(b) C_2H_2
(c) C_4H_{10}
(d) C_3H_6
Answer: (b)

Question: Consider the following reaction
$Ca + 2HCl \rightarrow CaCl_2 + H_2$
We have 14 g Ca reacts with excess of HCl. Choose the incorrect option.
Options:
(a) Mass produced of $CaCl_2$ is 38.85 g
(b) Mole of H_2 produced is 0.35 mol
(c) Volume of H_2 produced at STP is 7.84 L
(d) Mass of $CaCl_2$ product is 3.885 g
Answer: (d)

Question: Which one of the resonating structure is not correct
Options:
(a) 
(b) 

(c) 

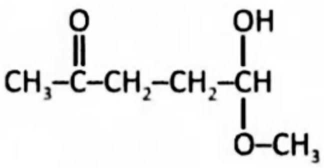
(d) 

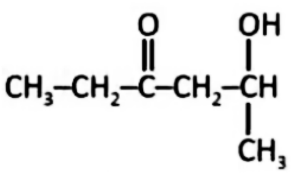
Answer: (c)

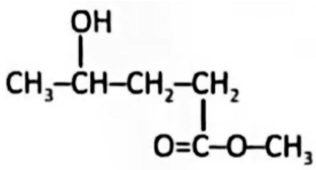
Question: Given below are two statements:
Statement-I: All the pairs of molecules (PbO, PbO_2); (SnO, SnO_2) and (GeO, GeO_2) contain amphoteric oxides.
Statement-II: $AlCl_3$, BH_3, BeH_2 and NO_2 all have incomplete octet.
In the light of the above statements, choose the correct option.
Options:
(a) Both Statement-I and Statement-II are correct
(b) Both Statement-I and Statement-II are incorrect
(c) Statement-I is correct but Statement-II is incorrect
(d) Statement-I is incorrect but Statement-II is correct
Answer: (d)

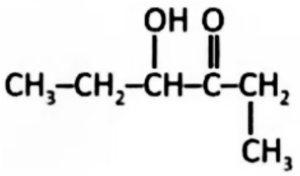
Question: Statement A: But-2-ene show O.I.
Statement B: Propanal and propanone are F.G.I
Statement C: Pentane and 2, 2-dimethylpropane are C.I.
Correct Statement are:
Options:
(a) Only A and B
(b) Only A and C
(c) Only B and C
(d) All
Answer: (c)

Question: $C_6H_{12}O_3$ gives positive iodoform test on hydrolysis with dil. Acid product formed gives Tollen and iodoform test both. Find structure of $C_6H_{12}O_3$.
Options:

(a) 

(b) 

(c) 

(d) 

Answer: (a)

Question: Consider the following statements.

A. **Propanal and Propanone are functional isomers**
B. **Ethoxyethane and methoxypropane are metamers**
C. **But-2-ene shows optical isomerism**
D. **But-1-ene and BUt-2-ene are functional isomers**
E. **Pentane and 2, 2-dimethylpropane are chain isomers**

The correct statement are

Options:

(a) A, B, D only
(b) B, C, D only
(c) A, B, E only
(d) A, B, D, E only

Answer: (c)

Question: Given below are two statements

Statement I: When electric discharge is put on hydrogen, it emits discrete frequency in electromagnetic spectrum.

Statement II: Frequency of He^{+} ion of 2^{nd} line of Balmer series is equal to first line of Lyman series of H-atom

Options:

(a) Both Statement I and Statement II are correct
(b) Both Statement I and Statement II are incorrect
(c) Statement I is correct and statement II is incorrect
(d) Statement I is incorrect and statement II is correct

Answer: (a)

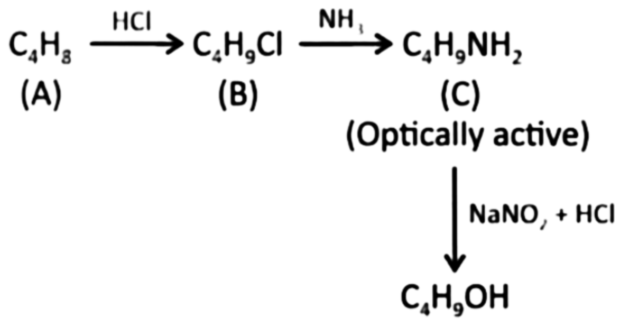
Question: Which of the following compound is paramagnetic in nature?

Options:

(a) $[Ni(CO)_4]$
(b) $[Ni(CN)_4]^{2-}$
(c) $[NiCl_4]^{2-}$
(d) $[Co(H_2O)_6]^{3+}$

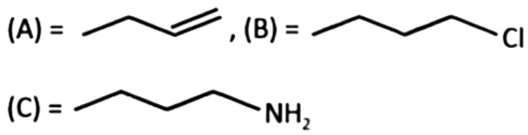
Answer: (c)

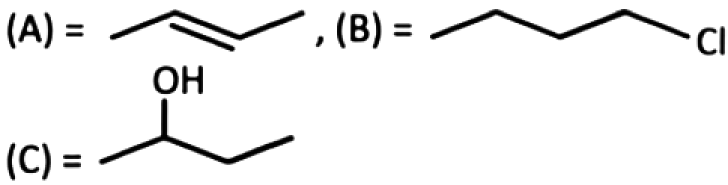
Question: Observe the following reaction sequence:

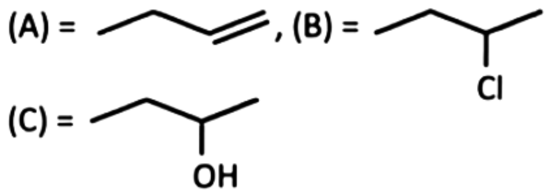


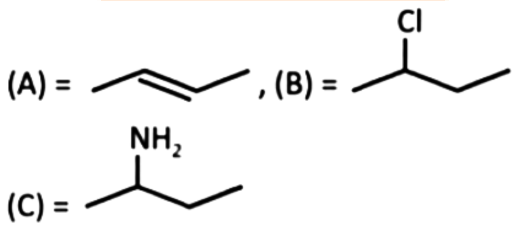
Which of the following is correct structure of A, B and C?

Options:

(a)


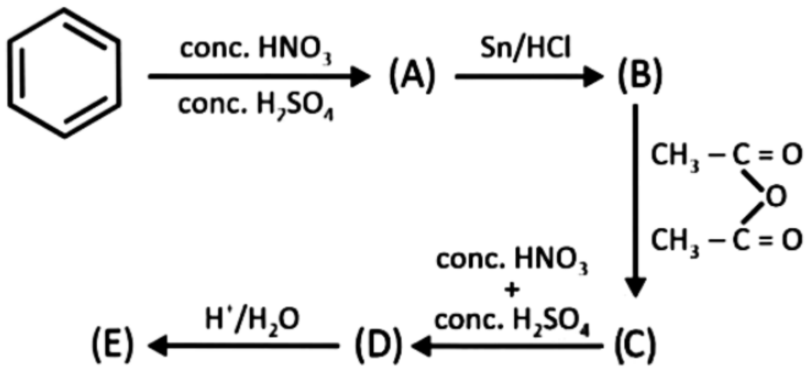
(b)


(c)


(d)


Answer: (d)

Question:



% of N in E = ?

Options:

Answer: (20)

Vedantu

Question: 10 mol of oxygen is heated at constant volume from 30ºC to 40°C. The change in the internal energy of gas is____cal. The molar specific heat of oxygen at constant pressure C_P = 7 cal/mol ºC & R = 2 cal/mol ºC
Options:
Answer: (500)

Question: 1 g of AB_2 is dissolved in 50 g solvent such that ΔT_f = 0.689. When 1 g AB is dissolved in 50 g of same solvent ΔT_f is 1.176. Find molar mass of AB_2. K_f = 5 K kg/mol. (Report to nearest integer) AB_2 and AB are non electrolyte.
Options:
Answer: (145)

Question: Out of the following, how many compounds have tetrahedral geometry?

$NH^{+}_{4}, XeF_{4}, \left[NiCl_{4}\right]^{2-}, \left[PtCl_{4}\right]^{2-}, \left[Cu\left(NH_{3}\right)_{4}\right]^{2+}, BF_{3}$ and $\left[Ni(Co)_{4}\right]$

Options:
Answer: (3)

JEE-Main-21-01-2026 (Memory Based)
[MORNING SHIFT]
Maths

Question:cosec 10° - $\sqrt{3}$ sec 10°
Options:
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (d)

$$\operatorname{cosec}10^o - \sqrt{3}\sec 10^o = \frac{1}{\sin 10^o} - \frac{\sqrt{3}}{\cos 10^o}$$

$$= \frac{\cos 10^o - \sqrt{3}\sin 10^o}{\sin 10^o \cos 10^o}$$

$$= \frac{2\left(\frac{1}{2}\cos 10^o - \frac{\sqrt{3}}{2}\sin 10^o\right)}{\frac{1}{2}\cdot\sin 20^o}$$

$$= \frac{4\cos 70^o}{\sin 20^o} = 4$$

Question: If $A = \begin{bmatrix}\alpha & 2\\ 1 & 2\end{bmatrix}, B = \begin{bmatrix}1 & 1\\ \beta & 1\end{bmatrix}$ **and A^2-4A + 2I = 0; B^2-2B + I = 0, then |adj(A^3- B^3)| is equal to**
Options:
(a) 11
(b) 7
(c) -11
(d) 121
Answer: (a)

$tr(A) = 4 \rightarrow \alpha = 2$

$|B| = 1 \rightarrow \beta = 0$

$A^3 = \begin{bmatrix}6 & 8\\ 4 & 6\end{bmatrix}\begin{bmatrix}2 & 2\\ 1 & 2\end{bmatrix} = \begin{bmatrix}20 & 28\\ 14 & 20\end{bmatrix}$

$B^3 = \begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}$

$\left|adj\left(A^3 - B^3\right)\right| = \left|A^3 - B^3\right| = \begin{vmatrix}19 & 25\\ 14 & 19\end{vmatrix}$

$= 361 - 350 = 11$

Question: If α and β are the roots of the equations $x^2 + x + 1 = 0$, then find $(\alpha+\beta)^4 + (\alpha^2 + \beta^2)^4 + (\alpha^3 + \beta^3)^4 + \ldots + (\alpha^{25} + \beta^{25})^4$

Options:
(a) 145
(b) 146
(c) 147
(d) 148
Answer: (a)

$$x^2 + x + 1 = 0$$

$$x = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$x = \frac{-1 \pm \sqrt{3}}{2}$$

$$x^2 + x + 1 = 0 \begin{array}{l} \nearrow \omega \\ \searrow \omega^2 \end{array}$$

$$(\alpha + \beta)^4 = \left(\omega + \omega^2\right)^4 = 1$$

$$\left(\alpha^2 + \beta^2\right)^4 = \left(\omega^2 + \omega\right)^4 = 1$$

$$\left(\alpha^3 + \beta^3\right)^4 = \left(\omega^3 + \omega^6\right)^4 = 16$$

$$16 \times 8 + 1$$

$$= 145$$

Question: The value of $\int_0^{\frac{\pi}{2}} |\sin x + \sin 2x + \sin 3x| dx$ **is**
Options:
(a) $\frac{8}{3}$
(b) $\frac{7}{3}$
(c) $\frac{2}{3}$
(d) 3
Answer: (b)

$$\int_0^{\frac{\pi}{2}} |2 \sin 2x \cos x + \sin 2x| \, dx$$

$$= \int_0^{\frac{\pi}{2}} | \sin 2x(2 \cos x + 1)|$$

$$I = \int_0^{\frac{\pi}{2}} \sin x + \sin 2x + \sin 3x \, dx$$

$$= -\cos x + \frac{-\cos 2x}{+2} + \frac{-\cos 3x}{+3} \Big|_0^{\frac{\pi}{2}}$$

$$= 1 + \left[\frac{1+1}{2}\right] + \left[\frac{0+1}{3}\right] = 1 + 1 + \frac{1}{3} = \frac{7}{3}$$

Question: **If y = y(x) and $(1+x^2)$ dy + (1- $\tan^{-1}x$)dx = 0 and y(0) = 1 then y(1) is**
Options:

(a) $\frac{\pi^2}{32}+\frac{\pi}{4}+1$

(b) $\frac{\pi^2}{32}-\frac{\pi}{4}+1$

(c) $\frac{\pi^2}{32}-\frac{\pi}{2}-1$

(d) $\frac{\pi^2}{32}-\frac{\pi}{2}+1$

Answer: (b)

$$dy+\frac{1+\tan^{-1}x}{1+x^2}dx=0$$

$$y+\tan^{-1}x-\frac{1}{2}\left(\tan^{-1}x\right)^2=c=1$$

$$y(1)+\frac{\pi}{4}-\frac{x^2}{32}=1$$

$$y(1)=1+\frac{\pi^2}{32}-\frac{\pi}{4}$$

Question: **The sum of roots of the equation $|x-1|^2$ -5 |x-1| + 6 = 0 is**
Options:

(a) 4

(b) 3

(c) 1

(d) 5

Answer: (a)

Let $|x-1|^2-5|x-1|+6=0$

Put $|x-1|=t$. Then

$t^2-5t+6=0$

$(t-2)(t-3)=0 \Rightarrow t=2 \text{ or } t=3$

So,

$|x-1|=2 \Rightarrow x=3,-1$

$|x-1|=3 \Rightarrow x=4,\ -2$

Sum of roots:

$3+(-1)+4+(-2)=\boxed{4}$

Question: **If L_1 and L_2 are two parallel lines and ΔABC is an equilateral triangle then area of triangle ABC is**
Options:

(a) $7\sqrt{3}$

(b) $4\sqrt{3}$

(c) $21\sqrt{3}$

(d) 84

Answer: (c)

$\sin\theta = \frac{6}{a}$

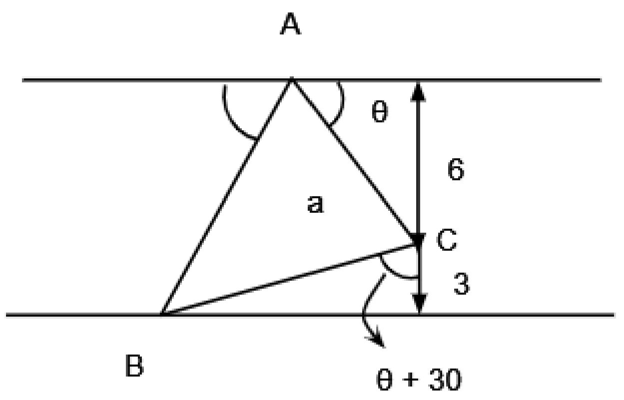
$\rightarrow \cos . \frac{\sqrt{3}}{2} - \sin\theta . \frac{1}{2} = \frac{3}{a}$

$\rightarrow \sqrt{1 - \frac{6^2}{a^2}} \times \frac{\sqrt{3}}{2} = \frac{3}{a} + \frac{3}{a} = \frac{6}{a}$

$1 - \frac{36}{a^2} = \frac{36}{a^2} \times \frac{4}{3} = \frac{48}{a^2}$

$1 = \frac{84}{a^2} \rightarrow a^2 = 84$

Area $= \frac{\sqrt{3}}{4} \times 84 = 21\sqrt{3}$



Question: The Hyperbola and ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ have same foci and eccentricity of Hyperbola is 5 then length of latus rectum of hyperbola is

Options:

(a) $\frac{96}{\sqrt{5}}$

(b) $24\sqrt{5}$

(c) $18\sqrt{5}$

(d) $12\sqrt{5}$

Answer: (a)

Ellipse:

$\frac{x^2}{36} + \frac{y^2}{16} = 1 \Rightarrow c^2 = 36 - 16 = 20 \Rightarrow c = 2\sqrt{5}$

Hyperbola has same foci and e = 5:

$a = \frac{c}{e} = \frac{2\sqrt{5}}{5}, a^2 = \frac{4}{5}$

For hyperbola:

$c^2 = a^2 + b^2 \Rightarrow b^2 = 20 - \frac{4}{5} = \frac{96}{5}$

Length of latus rectum:

$\mathrm{LLR} = \frac{2b^2}{a} = \frac{2.\frac{96}{5}}{\frac{2\sqrt{5}}{5}} = \boxed{\frac{96\sqrt{5}}{5}}$

Question: The number of relations, defined on the set {a, b, c, d} which are both reflexive & symmetric is equal to
Options:
(a) 16
(b) 1024
(c) 64
(d) 256
Answer: (c)
No of relations both reflexive and Symmetric = 26 = 64

Question: The locus of point of intersection of tangent drawn to the circle $(x - 2)^2 + (y - 3)^2 = 16$, which subtends an angle of 120° is
Options:
(a) $3x^2 + 3y^2 - 12x - 18y - 25 = 0$
(b) $x^2 + y^2 - 12x - 18y - 25 = 0$
(c) $3x^2 + 3y^2 + 12x + 18y - 25 = 0$
(d) $x^2 + y^2 + 12x + 18y$
Answer: (a)
For the circle
$(x-2)^2 + (y-3)^2 = 16$
the centre is (2,3) and radius r = 4.
If tangents drawn from a point P subtend an angle of 120°, then
$2\sin^{-1}\left(\frac{r}{OP}\right) = 120^\circ$
$\sin^{-1}\left(\frac{r}{OP}\right) = 60^\circ \Rightarrow \frac{r}{OP} = \frac{\sqrt{3}}{2}$
Hence,
$\mathrm{OP} = \frac{2r}{\sqrt{3}} = \frac{8}{\sqrt{3}}$
Therefore, the locus of point P is a circle with centre(2,3)and radius $\frac{8}{\sqrt{3}}$.
$$\boxed{(x-2)^2 + (y-3)^2 = \frac{64}{3}}$$
$$3(x-2)^2 + 3(y-3)^2 = 64$$
$$3\left(x^2 - 4x + 4\right) + 3\left(y^2 - 6y + 9\right) = 64$$
$$\boxed{3x^2 + 3y^2 - 12x - 18y - 25 = 0}$$

Question: If $a_1, a_2, a_3, \ldots$ are in increasing geometric progression such that $a_1 + a_3 + a_5 = 21$,
$a_1a_3a_5 = 64$ then $a_1 + a_2 + a_3$ is
Options:
(a) 5
(b) 7
(c) 10
(d) 15
Answer: (b)

$$a\,.\,ar^2.\;ar^4 = 64$$

$$\left(ar^2\right)^3 = 64$$

$$ar^2 = 4$$

$$a + ar^2 + ar^4 = 21$$

$$a + ar^4 = 17$$

$$a + a.\,\frac{16}{a^2} = 17$$

$$a + \frac{16}{a} = 17 \Rightarrow a = 1,\, r = 2$$

$$a_1 + a_2 + a_3 = 1 + 2 + 4 = 7$$

Question: In the expansion $(ax^2 + bx + c)(1 - 2x)^{26}$, if coefficient of x, x^2 & x^3 are -56, 0, 0 respectively, then find the value of a + b + c.

Options:

(a) 1500

(b) 1403

(c) 1300

(d) 1483

Answer: (b)

Coeff of x = -56

Coeff of x^2 = 0

Coeff of x^3 = 0

By applying binomial theorem on Given expression, we have

b $^{26}C_0$ + c $^{26}C_1(-2)$ = -56 ⇒ b- 52 C = - 56 …..(1)

a $^{26}C_0$ + $^{26}C_1(-2)$b + $^{26}C_2(-2)^2$ c = 0 ⇒ a - 52 b + 1300 c = 0 ……..(2)

c $^{26}C_3(-2)^3$ + b $^{26}C_2$ $(-2)^2$ + a $^{26}C_1(-2)$ = 0 ⇒ a - 25 b + 400 C = 0 …..(3)

By (1) , (2) & (3)

$$\begin{bmatrix} 0 & 1 & -52 \\ 1 & -52 & 1300 \\ 1 & -25 & 400 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 0 & 1 & -52 & -56 \\ 1 & -52 & 1300 & 0 \\ 1 & -25 & 400 & 0 \end{array}\right]$$

By applying row operations

(1) $R_1 \leftrightarrow R_2$

(2) $R_3 \leftrightarrow R_3 - R_1$

(3) $R_3 \rightarrow R_3 - 27R_2$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & -52 & 1300 & 0 \\ 0 & 1 & -52 & -56 \\ 0 & 0 & 504 & 1512 \end{array}\right]$$

a = 1300

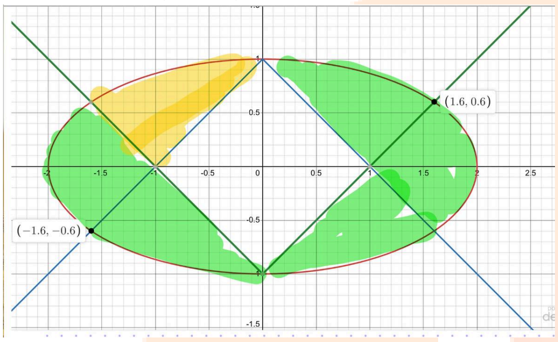
b = 100

c = 3

a+b+c = 1403

Question: Find the area inside the ellipse $x^2 + 4y^2 = 4$ and outside the area inclosed by the curves $y = |x| - 1$ & $y = 1 - |x|$.
Options:
(a) $2(\pi - 1)$
(b) $3(\pi - 1)$
(c) $\pi - 1$
(d) $\dfrac{\pi - 1}{4}$

Answer: (a)
Shaded Area = $\pi(2)(1) - (\sqrt{2})^2$
$= 2\pi - 2 = 2(\pi - 1)$



Question: The value of the integral $\displaystyle\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{2\pi + 4x^{11}}{1 - \sin\left(|x| + \frac{\pi}{6}\right)} dx$
Options:
(a) π
(b) 2π
(c) 4π
(d) 8π
Answer: (d)

$$\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{2\pi + 4x^{11}}{1 - \sin\left(|x| + \frac{\pi}{6}\right)} dx$$

$$I = 2\int_{0}^{\frac{\pi}{6}} \frac{2\pi\, dx}{1 - \sin\left(|x| + \frac{\pi}{6}\right)} \text{ (second part is odd)}$$

$t = x + \frac{\pi}{6}$

$$I = 4\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dt \times (1 + \sin t)}{(1 - \sin t)(1 + \sin t)}$$

$$I = 4\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\sec^2 t + \sec t \tan t\right) dt$$

$$I = 4\pi \left[\tan t + \sec t\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$I = 4\pi \times 2 = 8\pi$$

Question: If A={1, 2, 3, 4, 5, 6), B = {1, 2, 3, ... 8, 9}. Then the number of strictly increasing functions from A → B such that f(i) $\neq$ i $\forall$ i = 1, 2, 3, 4, 5, 6 are
Options:
(a) 25
(b) 27
(c) 28
(d) 29
Answer: (c)
Case I : f(1) = 2
No. of functions = $^7C_5$
= 21
Case II : If f(1) = 3
No of functions = $^6C_5$ = 6
Case III : If f(1) = 4
No of functions = $^6C_6$ = 1
Total = 21 + 6 + 1 = 28

Question: If the mean and variance of observations x, y, 12, 14, 4, 10, 2 is 8 and 16 respectively where x > y. Then, the value of 3x - y is
Options:
(a) 18
(b) 20
(c) 22
(d) 24
Answer: (a)

$\overline{X} = \frac{x+y+12+14+4+10+2}{7}$

$8 = \frac{x+y+42}{7}$

$x + y = 56 - 42$

$x + y = 14$

$a^2 = \overline{x^2} - (\overline{x})^2$

$16 = \frac{x^2+y^2 144+196+16+100+4}{7} - (8)^2$

$(16 + 64) \times 7 = x^2 + y^2 + 460$

$560 - 460 - x^2 + y^2$

$x^2 + y^2 = 100$

$(x+y)^2 = 2xy = 100$

$(14)^2 - 2xy = 100$

$xy = 48$

$x + y = 14 \quad xy = 48$

$x + \frac{48}{x} \ = 14$

$x^2 - 14x + 48 = 0$

$x = 8, 6$

$y = 8, 6$

$x > y = 1\,,\ x = 8_1\,,\ y = 6$

$3x - y = 3 \times 8 - 6 = 18$

Question:If O is the vertex of the parabola $x^2 = 4y$, Q is the point on the parabola. If C is the locus of a point which divides OQ ratio 2:3. The equation of the chord of C which bisected at point (1, 2).

Options:

(a) 5x + 4y + 3 = 0

(b) 5x - 4y - 3 = 0

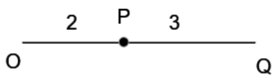
(c) 5x - 4y + 3 = 0

(d) 5x + 4y - 3 = 0

Answer: (c)

$x^2 = 4y$

$0(0,0) \quad Q(2t_1\, t^2)$



$P = \left(\frac{4t}{5}, \frac{2t^2}{5}\right)$

$x = \frac{4t}{5},\ y = \frac{2t^2}{5}$

$t = \frac{5x}{4},\ y = \frac{2}{5}\left(\frac{5x}{4}\right)^2$

$y = \frac{5}{8}\,x^2$

$T = s_1 \ \rightarrow \ (1,\ 2)$

$\frac{y+2}{2} \ - \frac{5}{8}\,x(1) \ = 2 - \frac{5}{8}$

$5x - 4y + 3 = 0$

Question: If f(3) = 18, f'(3) = 0 and f''(3) = 4. Then, the value of

$$\lim_{x \to 1} \ell n \left(\frac{f(x+2)}{f(3)} \right)^{\frac{18}{(x-1)^2}}$$

is equal to

Options:

(a) 2

(b) 4

(c) 6

(d) 8

Answer: (a)

$$L=\log_e\left(\lim_{x\to 1}\left(\frac{f(x+2)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right)$$

1^{∞}form

$$L=\log_e\left(e^{\lim\limits_{x\to 1}\frac{18}{(x-1)^2}\left(\frac{f(x+2)}{f(3)}-1\right)}\right)$$

$$L=\log_e\left(e^{\lim\limits_{x\to 1}\frac{18}{(x-1)^2}\left(\frac{f(x+2)-f(3)}{f(3)}\right)}\right)$$

$$L=\log_e\left(e^{\lim\limits_{x\to 1}\left[\frac{18(f(x+2)-18)}{(x-1)^2\,18}\right]}\right)$$

$$L=\log_e\left(e^{\lim\limits_{x\to 1}\left[\frac{f(x+2)-18}{(x-1)^2}\right]}\right)$$

$$L=\log_e\left(e^{\lim\limits_{x\to 1}\frac{f'(x+2)}{2(x-1)}}\right)$$

$$L=\log_e\left(e^{\lim\limits_{x\to 1}\frac{f''(x+2)}{2}}\right)$$

$$L=\log_e\left(e^2\right)=2$$

Question: If $a_1=1$ and for $n\geq 1$, $a_{n+1}=\frac{1}{2}a_n\frac{n^2-2n-1}{n^2(n+1)^2}$ **then** $\left|\sum_{n=1}^{\infty}\left(a_n-\frac{2}{n^2}\right)\right|$

Options:

(a) 4

(b) 2

(c) 1

(d) 5

Answer: (b)

$$a_{n+1}=\frac{a_n}{2}+\frac{1}{(n+1)^2}-\left[\frac{(n+1)^2-n^2}{n^2(n+1)^2}\right]$$

$$a_{n+1}=\frac{a_n}{2}+\frac{1}{(n+1)^2}-\left[\frac{1}{n^2}-\frac{1}{(n+1)^2}\right]$$

$$a_{n+1}-\frac{2}{(n+1)^2}=\frac{a_n}{2}-\frac{1}{n^2}$$

$$a_{n+1}-\frac{2}{(n+1)^2}=\frac{1}{2}\left(a_n-\frac{2}{n^2}\right)$$

$$b_{n+1}=\frac{1}{2}b_n \qquad b_n=a_n-\frac{2}{n^2}$$

$$b_2=\frac{1}{2}(-1)=-\frac{1}{2} \qquad b_1=1-\frac{2}{1}=-1$$

$$b_3=\frac{1}{2}\left(\frac{-1}{2}\right)=-\frac{1}{4}$$

$$b_n=(-1)\left(\frac{1}{2}\right)^{n-1}=-\left(\frac{1}{2}\right)^{n-1}$$

$$a_n-\frac{2}{n^2}=-\left(\frac{1}{2}\right)^{n-1}$$

$$\left|\sum_{n=1}^{\infty}\left(a_n-\frac{2}{n^2}\right)\right|=\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^{n-1}$$

$$=1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots..+\infty$$

$$=\frac{1}{1-\frac{1}{2}}=2$$

Question: If the domain of the function

$$\cos^{-1}\left(\frac{2x-5}{11x-7}\right)+\sin^{-1}(2x^2-3x+1) \text{ is } [0,\, a],\ \left[\frac{12}{13},\, b\right] \text{ then } \frac{1}{ab}$$

is equal to

Options:

(a) -3

(b) 3

(c) 2

(d) 4

Answer: (b)

$$-1\le\frac{2x-5}{11x-7}\le 1, \qquad -1\le 2x^2-3x+1\le 1$$

$$\frac{2x-5}{11x-7}+1\ge 0,\ \frac{2x-5}{11x-7}-1\le 0,\ 2x^2-3x+2\ge 0,$$

$$2x^2-3x\le 0$$

$$\frac{13x-12}{11x-7}\ge 0,\ \frac{-9x+2}{11x-7}\le 0,\ x(2x-3)\le 0$$

$$x\in\left(-\infty,\frac{7}{11}\right)\cup\left[\frac{12}{13},\infty\right),\ x\in\left(-\infty,\frac{2}{9}\right]\cup\left(\frac{7}{11},\infty\right),\ x\in\left[0,\frac{3}{2}\right]$$

$$\Rightarrow x\in\left[\frac{12}{13},\frac{3}{2}\right]\cup\left[0,\frac{2}{9}\right]$$

$$a=\frac{2}{9},\ b=\frac{3}{2}\Rightarrow ab=\frac{1}{3}\Rightarrow\frac{1}{ab}=3$$